You can still see the messages through the *dmesg* command. Disabling the console output comes under the category of 'Don't do it at all'—see whether you can postpone an activity done during the boot-up.

Delay the loading of drivers

The loading of a few drivers can be delayed and this can also considerably reduce the boot-up time. For example, certain drivers can be loaded later in the boot sequence or in parallel with other drivers or applications. This comes under the category of 'Do it Later' [2]—see whether you can postpone an activity done during the boot-up [6].

Smaller kernel

The Linux kernel will load faster if it is lean and trim. In that sense, the CELF's specification on size reduction, indirectly contributes to boot-up time reduction also.

The Linux kernel is designed for servers and desktops. Embedded systems are dedicated systems and hence they do not require many features that are applicable for servers. During kernel configuration, we can disable kernel features not required, which results in kernel size reduction. Further information on having a smaller kernel is provided in the second part of this article.

An excellent instant loading technique

Boot up the kernel, start all the applications and the user interface, then suspend to disk or flash in the same state. When the device is booted again, the system state is restored from this predefined hibernation image. This technique is followed in the Sony Digital Still Camera and a few handheld devices. With this technique, it is possible to achieve instant 'Power On' in less than 400 msec [3].

Power management

One USP (unique selling proposition) consumer electronics device manufacturers certainly need is low power consumption. Power management has gained significance of late because of the importance given to longer battery life in handheld devices and mobile phones, and also because of the campaign to turn green. What is the cause of higher power consumption in CE devices?

Well, if you were to dig deep into the roots of devices and circuits, you will find that power consumption in embedded devices depends on three factors.

1. Leakage current power dissipation
2. Dynamic current power dissipation
3. Short circuit current power dissipation

Leakage current power dissipation is due to the flow of leakage current and this is dependent on operating voltage [1].

Dynamic current power dissipation is due to the flow of dynamic current and this is the most important form of dissipation. This power dissipation will occur when the device is active, and it is dependent on the operating voltage and the frequency of input.

Short circuit current power dissipation is due to the flow of short circuit current and it is also dependent on voltage.

Figure 3: Option to enable frequency scaling during kernel configuration

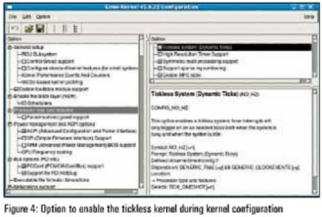

Figure 4: Option to enable the tickless kernel during kernel configuration

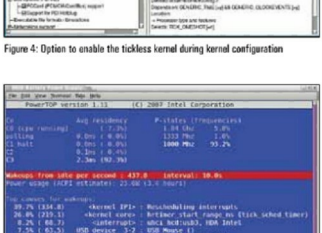

Figure 5: Power Top output when a laptop runs on battery power supply

Among the three forms of power dissipation, only dynamic current power dissipation is proportional to voltage and frequency ([1] and [6]).

The principles of power management are: